Who is hiding the X-rays?

Unveiling the X-ray emission of AGNs

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1 The disc-corona connection

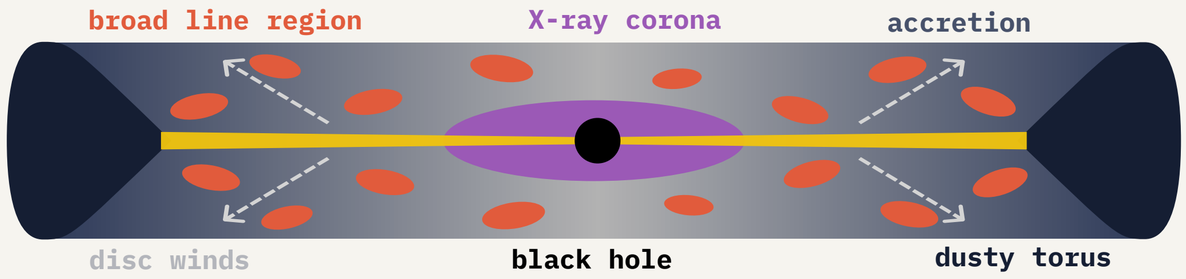


Fig. 1 AGN unification scheme.

In active galactic nuclei (AGNs), the accretion disc around a supermassive black hole (BH) emits thermal UV radiation. Some of the photons is up-scattered to X-ray energies on the relativistic free electrons of X-ray corona. Such X-ray emission can be modeled as a powerlaw [1]. The accretion disc produces the black-body continuum, which is interwoven with spectral lines. Fig. 1 illustrates this AGN model.

The disc-corona coupling gives rise to the observed sub-linear relation between the X-ray luminosity and UV luminosity

$$\log L_X = 0.58 \log L_{UV} + 8.9$$

reffered to as the ***L_X-L_{UV}* relation** [2, 3]. With a dispersion reduced to ~0.2 dex, the relation is a probe of disc-corona physics and through its sub-linearity a way to measure luminosity distances of quasars out to z~7 [4].

2 X-ray weak AGNs

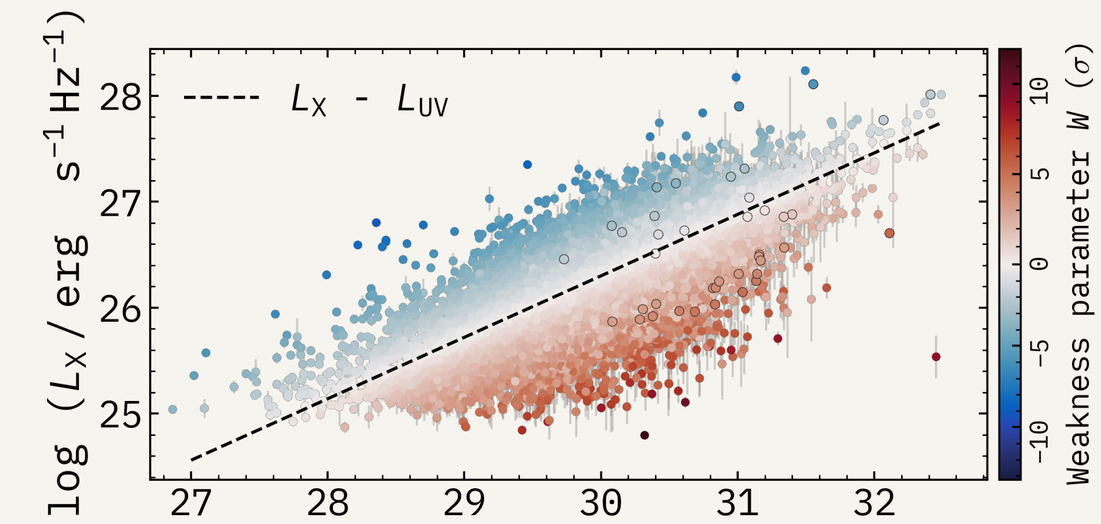


Fig. 2 LX-LUV plane for the initial quasar sample. X-ray weak AGNs lie below the relation.

A subset of quasars, lies significantly bellow the LX-LUV plane (Fig. 2). We quantify their deviation with the weakness parameter

$$W = 1 - \frac{\log L_{X,obs}}{\log L_{X,pred}}$$

in units of its uncertainity σ. Objects with W>0 are **X-ray weak AGNs**. They tend to show flatter X-ray spectra [5], and weaker, more blueshifted C IV emission. They are universal accros accretion rates and black hole masses [6].

3 Hidden X-ray emission?

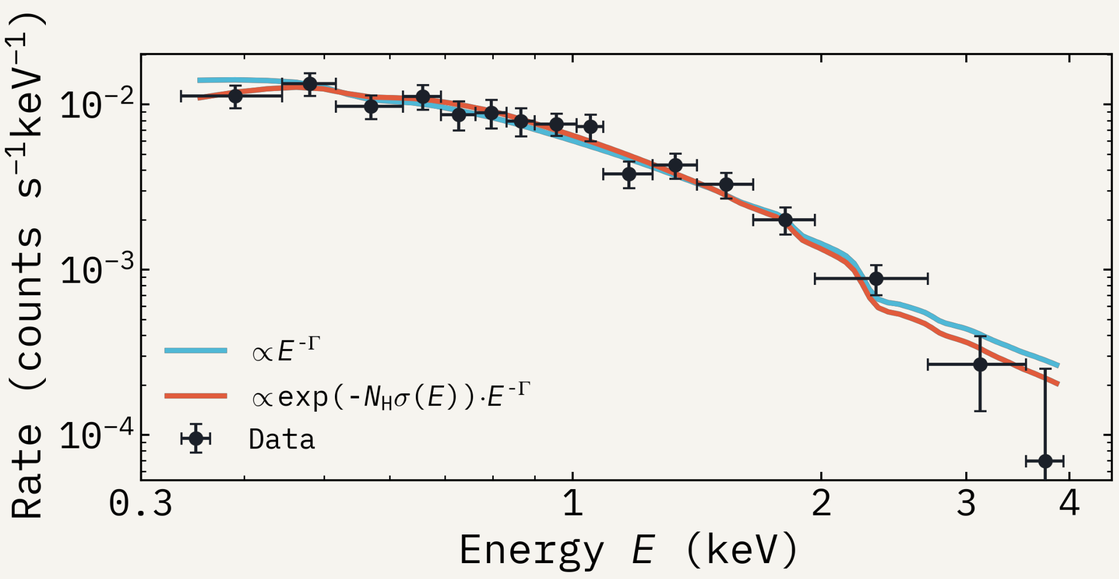


Fig. 3 X-ray spectrum of a quasar at z~2. Blue solid line shows unabsorbed powerlaw, red one shows powerlaw with absorption component.

X-ray weakness arises either from absorption of a normal corona or from an intrinsically under-luminous one. Firstly, photoelectric **absorption** flattens the spectrum, mimicking a low photon index Γ. Photometry cannot tell the two apart, but spectroscopy with enough counts reveals the absorption curvature and measures the absorbing column N_H.

Alternatively, the weakness is linked to **line-driven accretion disc winds** [5, 6]. Line driving requires a low X-ray flux in the launching region, so that the gas is not over-ionised [7]. Such winds can appear as blueshifted, asymmetric profiles of high-ionisation lines such as C IV.

4 Data & methods

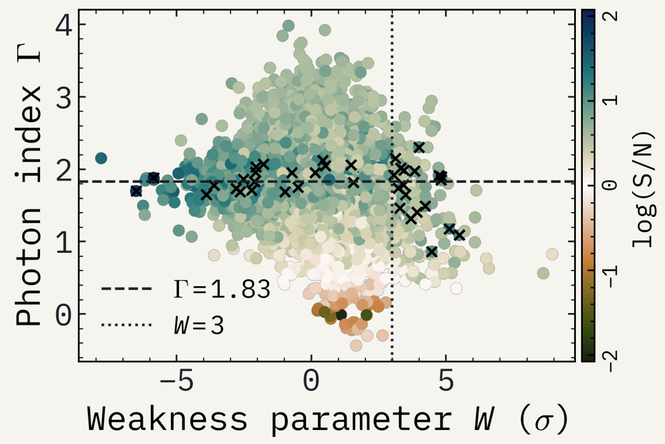


Fig. 4 Sample selection.

Our sample initially contained about 18,000 quasars, for which we had XMM-Newton photometry [8] and SDSS spectroscopy [9].

Archival **XMM-Newton** spectroscopic data of 40 selected objects were analyzed. Their choice is illustrated in Fig. 4.

For each spectrum we found the best-fit model, derived Γ, N_H if need be, and luminosity/flux at 2 keV.

5 Results

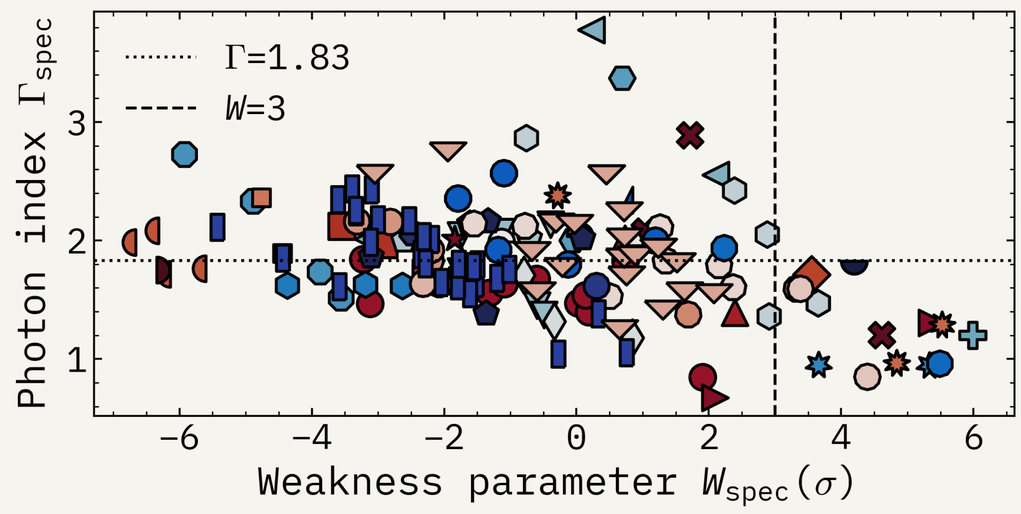


Fig. 5 Spectroscopic values of photon index against weakness parameter. A lack of X-ray weak AGNs with steep spectra observed.

Since the sample was split by half based on X-ray weak/normal, it is surprising to see that only **11%** of observations show X-ray weakness. Additionally, X-ray weak AGNs exhibit solely **flat spectra,** see Fig. 5.

X-ray weakness seems to be a **transient** state, see Fig. 6. Spectroscopy yields **systematically lower** values by 0.73σ on average when compared to photometric measurements, with more than 5σ significance.

Only 1/239 observations require absorption at more than 99% significance, meaning that **absorption isn't the primary** cause for the weakness. This suggests flat spectra to be an attribute of the X-ray weak AGNs.

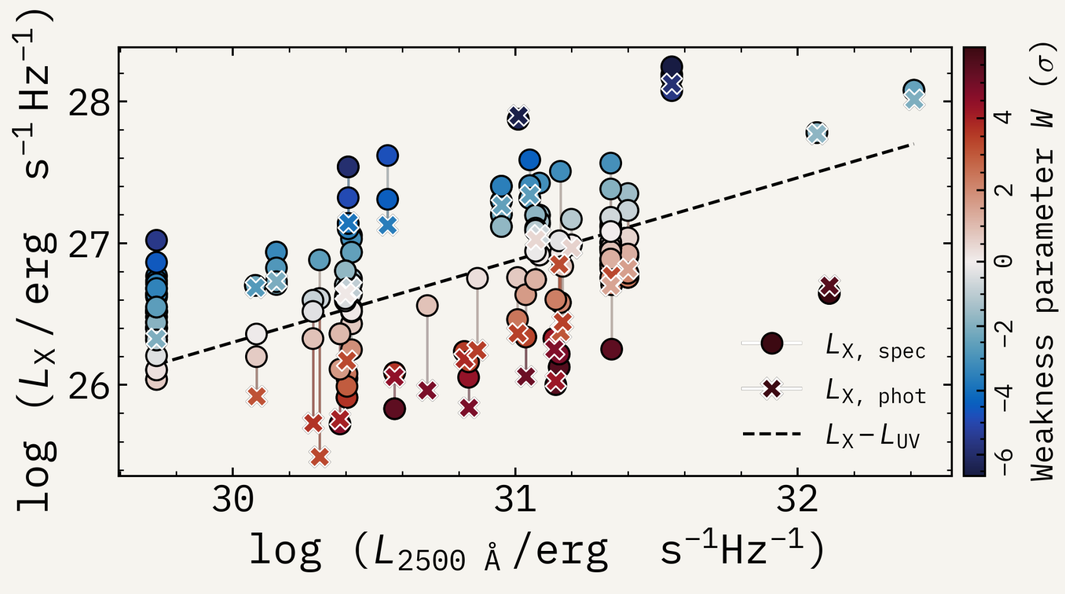


Fig. 6 Evolution of X-ray weakness for our sample ilustrated in the L_X-L_{UV} plane.

Take-home message

Geometry of AGNs reflects what we observe, with *L_X-L_{UV}* relation being the most striking example.

X-ray weak AGN contribute to the scatter in the *L_X-L_{UV}* relation. It seems that their weakness is overestimated in photometric studies, and may not be cause by absorption. Furthermore, X-ray weakness might be only a transient state of the AGNs.

Constraining the nature of X-ray weakness will allow us to use AGNs as standard candles to much higher redshifts.

References: [1] Haardt, F., & Maraschi, L. 1991, ApJ, 380, L51; [2] Avni, Y., & Tananbaum, H. 1986, ApJ, 305, 83; [3] Lusso, E., & Risaliti, G. 2016, ApJ, 819, 154; [4] Risaliti, G., & Lusso, E. 2019, NatAs, 3, 272; [5] Nardini, E., et al. 2019, A&A, 632, A109; [6] Trefoloni, F., et al. 2023, A&A, 677, A111; [7] Murray, N., et al. 1995, ApJ, 451, 498; [8] Webb, J. K., et al. 2020, A&A, 641, A136; [9] Wu, Q., & Shen, Y. 2022, ApJS, 263, 42

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